Fluent, Generic, and Unfaithful: Evaluating LLM Human Simulations Against Real Consumer Answers

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The Problem

A Beauty & Wellbeing company built LLM-based simulations of individual consumers from a single open-ended interview, following Park et al. (2024). Can those simulations replace real respondents' open-ended answers? We treat the evaluation as a *risk control*: where can simulated answers be trusted for decisions, and where not. Ground truth is the real humans' answers to a second interview (3 people × 10 questions, one human and one AI answer per pair).

Our Outcome

The simulations are **not reliably distinguishable in blind reading, generic rather than individual, and unfaithful** to what the real people said.

Not reliably distinguishable. Blind judges (one answer, no label, randomised order) discriminate human from AI at only 0.55 / 0.58 / 0.67 against 0.50 chance. All three judges could better identify human answers than AI (Figure 1) but could not agree reliably per item. Krippendorff's α (Krippendorff, 2004) is chance-corrected agreement among the three judges on identical items (1 perfect, 0 chance); here $\alpha = 0.396$, a weak-confidence result.

Generic. When asked whether the AI answer is recognisably *this* person rather than a generic segment member, the judges could not conclude deterministically.

Unfaithful. These numbers come from the deterministic calculations using encoders. Coverage (how much of what the person said the simulation kept) is 16.5%; fidelity (how much of what the simulation said the person backs) is 4.4%, with 3 outright contradictions across 30 pairs. The measure is strict by construction (sentence-level entailment), so absolute values are understated.

The combination of the three KPIs makes inference unreliable: detectability judges the *form* of one text alone; coverage and fidelity compare *content* against the person's real answer. Fluent in-domain text passes blind reading while its specifics are false, which is why a measurement layer is required.

Exploratory judgements agree. Three judge families each scored the same 12 held-out items (q7 to q10 for every person), so one verdict is one judge on one question. Personality capture: 7 of 36 verdicts yes (4, 2, 1 per judge). Content capture: 9 of 36 verdicts yes (4, 3, 2 per judge). Zero-shot human-or-AI accuracy is 0.58 to 0.78 per judge and asymmetric: every judge recognises humans better (0.77 to 0.90) than it catches simulations (0.30 to 0.67; Figure 1).

The style gap, measured. On the deterministic features of Figure 2, the AI writes 55 words to the human's 31, with far richer vocabulary (MTLD 95 vs. 56; McCarthy and Jarvis, 2010), hedges less (2.35 vs. 5.57 per 100 words), and is strongly positive where humans are near neutral (VADER mean 0.68 vs. 0.28; Hutto and Gilbert, 2014). The simulations sell; the humans shrug.

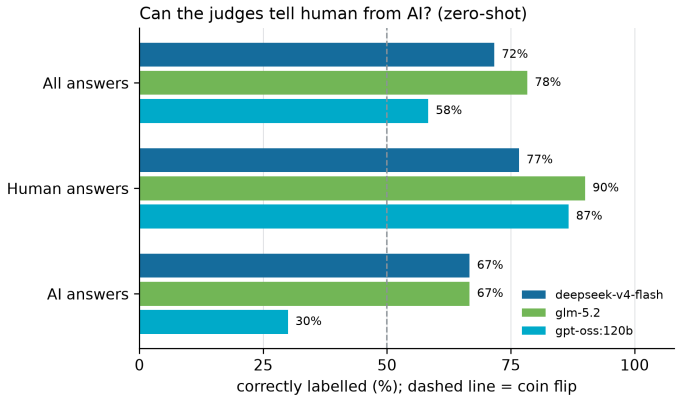


Figure 1: Zero-shot detection accuracy per judge model. Bars show the share of the 60 answers each judge labelled correctly, overall and by true side; the dashed line marks the 50% coin-flip level. All three judges recognise humans better than they catch simulations.

Methodology

We started from evaluation criteria where each criterion led to one metric.

(a) Does it say what the person said: content coverage and fidelity via NLI.

(b) Does it sound like a person, and like *this* person: style features plus an individuation judge.

(c) Could anyone tell: blind believability.

(d) Do the two sides talk about the same themes: a topic map.

Below is the detailed methodology with its strengths and weakness.

- **Method choice:** sentence-level NLI in both directions with a DeBERTa cross-encoder (He et al., 2021) (coverage and fidelity; SummaC lineage, Laban et al., 2022).
  Strength: deterministic, auditable, direction-aware; the two directions punish padding and thinness respectively.
  Weakness: strict entailment on conversational text understates absolute values (domain shift).
- **Method choice:** seven deterministic stylometric features per answer (length, vocabulary, hedging, sentiment; hedging lexicon from Hyland, 2005).
  Strength: fully deterministic, published lexicons, reproducible.
  Weakness: surface-level; cannot see meaning.
- **Method choice:** LLM judge panel of three model families (gpt-oss:120b, deepseek-v4-flash, glm-5.2) scoring identical items; judges output only booleans or bounded scores; aggregates computed in code.
  Strength: family diversity counters self-preference; head-to-head comparable; agreement measurable.
  Weakness: judges are themselves LLMs; no human raters; the believability construct is weak at $\alpha = 0.396$.
- **Method choice:** blind believability (one unlabelled answer, randomised order).
  Strength: blinding plus randomisation; the discrimination rate is computed in code.

*Prepared for Roland Berger.

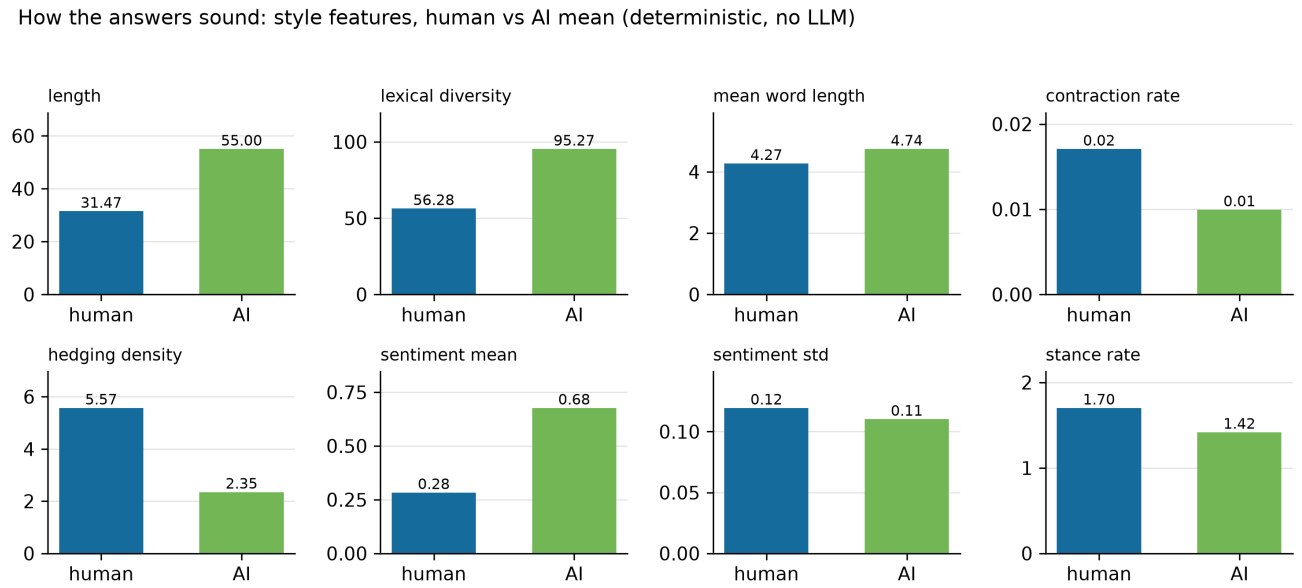


Figure 2: Deterministic style features, measured on every answer. Each panel compares the human mean (blue) with the AI mean (green) for one of the seven features, over 30 answers per side. The AI writes longer, more varied, less hedged and far more positive text than the people it simulates.

Weakness: low inter-judge agreement; judges biased toward calling text human.

- **Method choice:** individuation with a persona sketch (the person's other nine answers).
  Strength: the judge sees the person's own words; our most reliable result ($\alpha = 0.615$).
  Weakness: sketch quality bounded by nine short answers.
- **Method choice:** seeded BERTopic topic map over pooled human/AI sentences.
  Strength: targets the population-structure gap the reference paper concedes.
  Weakness: degenerate at this n; reported as not computable.
- **Method choice:** fixed explainable split (q7 to q10 held out for every person), lowercased judge inputs, full disk cache.
  Strength: explainable, reproducible; orthography confound removed. *Weakness:* lowercasing also removes any real stylistic case signal.

Caveats and Limitations

- $n = 30$ **pairs.** We could not test significance level clearly and had to aggregate at human level. This is because per-category slices were too small (3 to 15 pairs).
- **Orthography confound, found and corrected.** Run-1 personality verdicts keyed on lowercase and typos, transcription artefacts rather than voice; judge inputs were lowercased in subsequent runs
- **Topic-level seggregations not computable at this** n The first BERTopic fit collapsed into one stopword mega-topic. We corrected for stopwords and used mpnet embeddings with `n_neighbors` = 10). This still led to generic topics with no clear distinction. A similarity-graph alternative was rejected as it would be sensitive to similarity measure thresholds.
- **Judge layers are not stable** across cloud reruns despite temperature 0 and fixed seeds, while deterministic layers reproduce exactly.
- **No test-retest ceiling.** We could not test humans stability as they were normalised in Park et al. (2024).
- **Generation prompts unknown.** Style deviations may be prompt artefacts (chain-of-thought, expert reflections). These could not be tested

Recommendation

Use the simulations for ideation, hypothesis generation, and question piloting: tasks that need plausible consumer-shaped text.

Do not use hem for preference measurement, verbatim-style quotes, or individual-level insight.

The simulations cannot be trusted as substitutes for these consumers' answers. They are not reliably distinguishable from human answers in blind reading, which makes their failures invisible to a reader; they revert to a generic segment member rather than the individual; and almost nothing they say is backed by what the person actually said, including three outright contradictions. Their fluency is precisely what makes them dangerous: nothing warns the reader that the content is wrong. A distinct but supporting observation is that the LLM components of the evaluation itself, the judges, were not run-to-run stable despite temperature 0 and fixed seeds; if LLMs cannot judge consistently, relying on them to *be* the consumer is a larger leap. The stability of the simulations themselves was not testable with a single generation per question and isx' our first data request.

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